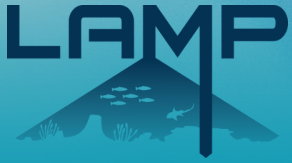


# User Guide



Welcome to iSEA (Intelligent Seafloor & Animal Image Annotator), a tool for annotating underwater images. This manual will guide you through the software's features and functionality, allowing you to make the most of its capabilities.

## Overview

iSEA is an application designed to facilitate annotation and object detection in images. Equipped with artificial intelligence, it enables automatic fauna detection and seafloor classification, offering tools for manual annotation, real-time annotation, and custom model training.

## Key features

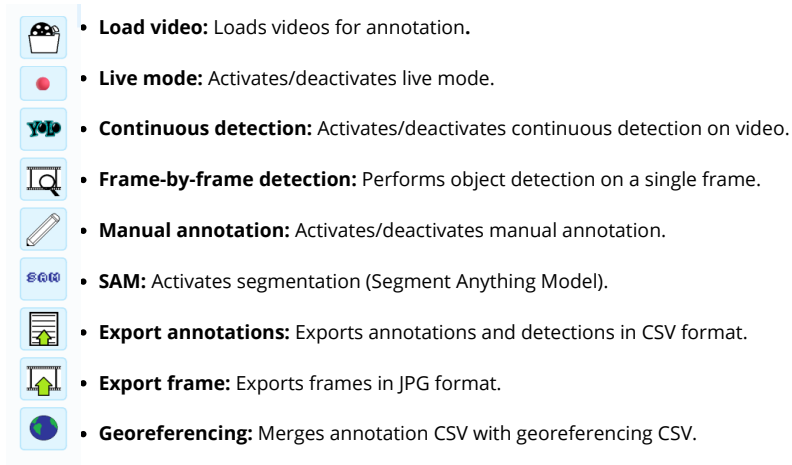
- **Automatic fauna detection:** Uses YOLO models to identify and track animals by video.
- **Manual annotation:** Provides tools for you to create your own annotations.
- **Live mode:** Connects to a camera for visualization and real time annotation.
- **Video recording:** Allows you to record videos directly from the live camera.
- **Model training:** Trains custom models using manual annotations and exports datasets in YOLO format.
- **Annotation export:** Exports annotations in CSV format.

## 1. User interface

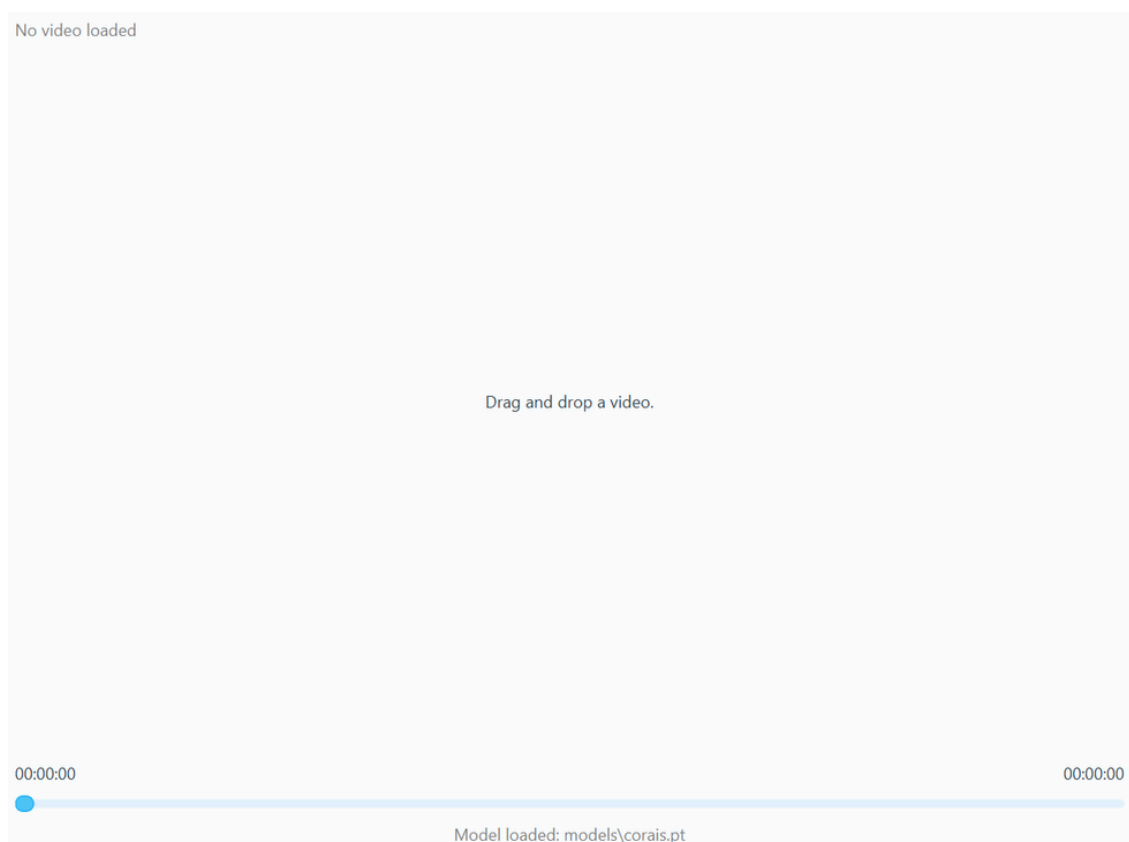
### Top menus

- **File:** Load video, load and unload models, and save annotations.
- **View:** Toggle between detection history view modes and activate dark mode.
- **Annotation:** Perform frame-by-frame detection, activate continuous detection, enable manual annotation and enrich the annotation taxonomy using WoRMS.
- **Train:** Create and export training datasets, train models.
- **Seafloor:** Manage seafloor categories, train classifier.
- **Language:** Change the interface language.
- **Help:** Display shortcuts and information about the software.

## Toolbar

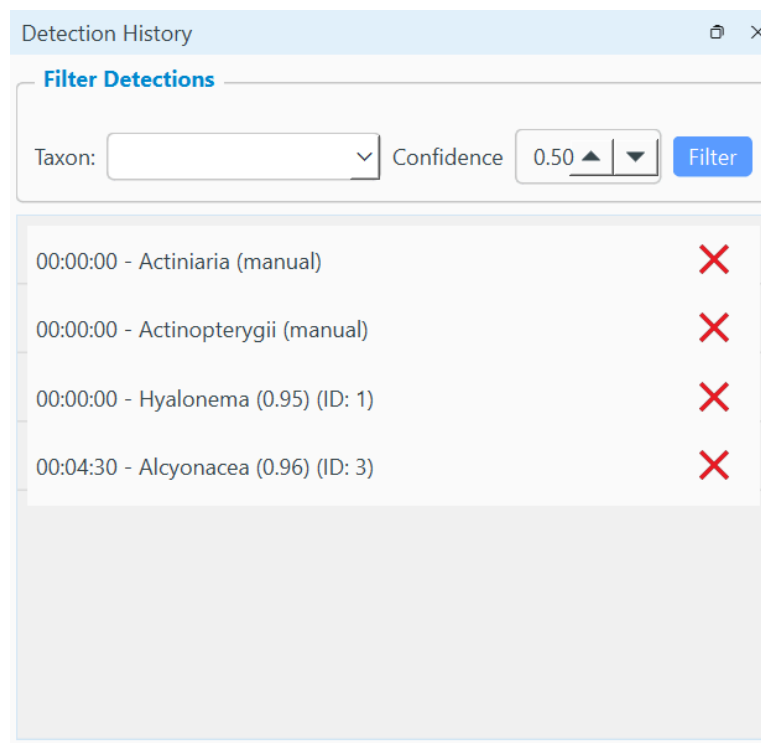


## Image display area



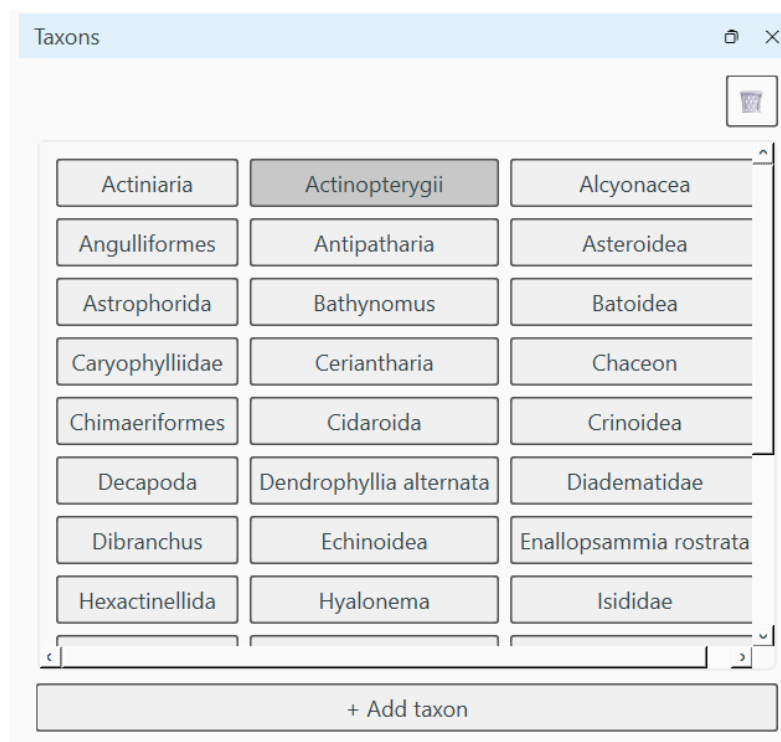
- Displays the loaded video or live camera feed.
- Allows viewing current annotations and interacting with them.
- Draggable playback menu at the bottom.
- 2x Speed:
  - Continuous detection disabled: doubles the video FPS.
  - Continuous detection enabled: doubles video FPS + detects at spaced intervals, discarding the current frame if it's similar to the previous frame.

## Detection history



Draggable box where detections and manual annotations appear, with option to filter by taxon or confidence level.

## Taxon grid



The grid starts with default classes from a coral bank trained YOLO model, in alphabetical order, and is automatically updated when a custom model is loaded. To annotate:

- Activate "Manual Annotation".
- Click on the desired taxon.

- c. Draw the box over the organism in the image (Try to make the box as small as possible, without leaving too much background around it.)

Delete taxons: click the  icon (top right corner), select the taxons to be removed, and confirm by clicking the trash again.

Add new taxon: use the "+ Add taxon" button at the bottom of the grid.

## 2. Manual annotation



When activating manual annotation mode, the user can select a taxon from the taxon grid (located in the lower right corner; see item 1, User Interface) and then draw the bounding box directly on the image, indicating the position of the corresponding organism.

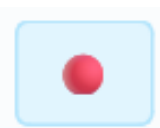
### 2.1. SAM (Segment Anything Model):



Enable SAM to create segmented annotations:

- a. Select a taxon from the taxon grid (see item 1, User Interface).
- b. Hover your mouse cursor over the target organism to see a segmentation suggestion from the system.
- c. Left-click to create the annotation.

## 3. Live mode



Activates the connected camera for real-time annotation/detection.

With live mode on, use File > Start recording if you want to save the video.

## 4. Automated detection



a. Continuous detection.

b. Detection by frame.

There are two options for automated detection: the continuous detection and the detection by frame.

### a. *Continuous detection*

Activates the loaded YOLO model, performing detection on all video frames continuously. However, if 2x speed is enabled, detection occurs at spaced intervals, discarding the current frame if it is similar to the previous frame.

### b. *Detection by frame*

Activates the loaded yolo model but only in the current frame, pausing the video to show detected taxa.

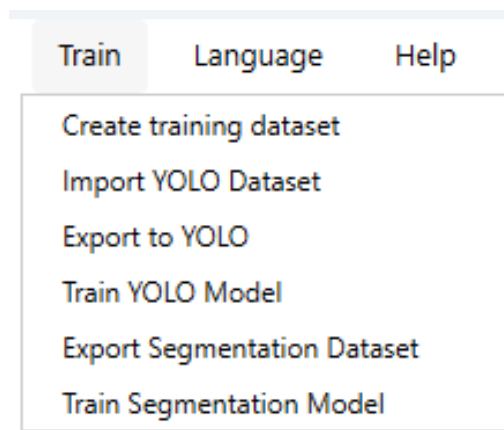
## 5. Seafloor classifier

Initially, the classifier has three categories: sediment, coral fragment, and coral reef. New categories can be added at any time via the *Seafloor* menu. The classifier operates in two ways:

- **Training with an image database:** Through the *Seabed* menu, the user loads a set of images and starts the classifier. The system prompts the selection of 10 images per category, using these annotations to train the model.
- **Training during video annotation:** During annotation, the user can classify the seafloor type using keyboard shortcuts: S for sediment, F for coral fragment, and R for coral reef. The video segment is annotated with the selected category until the user chooses another seafloor, and the corresponding frames are automatically saved for model training. The Shift+S key stops data collection, and the "Train with Collected Data" option generates the model from these samples.

With the model already trained and loaded, the system classifies the background every five frames during playback, displaying the result along with the confidence level (e.g., "Sediment: 0.92") in the interface's information panel, integrating seamlessly into the annotator's workflow.

## 1. Training mode



iSEA offers two flexible approaches for creating your own custom detection models:

### ***1. Complete approach: Structured dataset creation***

Ideal for new projects.

#### **Step by step:**

- a. **Start Dataset:** In the Training menu, select “Create Dataset.”
- b. **Select data source:**

**Option 1 - Image Set:** Select multiple images that have already been captured.

**Option 2 - Video extraction:** Load a video as a source. Set the Extraction Factor (e.g., 1 frame every 10 frames). The program automatically extracts frames at regular intervals.

- c. **Annotation:** Navigate between frames using the right/left arrows on your keyboard. Use the taxon grid to record all visible organisms. Continue until you have recorded all frames in the dataset.
- d. **Final Processing:** Export to YOLO (or export segmentation dataset): Saves the annotations (labels + images + dataset configuration file). Train YOLO Model (or segmentation model): Exports the annotations and automatically starts training the model.

#### **Advantages of this approach:**

- a. Organized and structured dataset since the beginning.
- b. Precise control over which frames are included.
- c. Ideal for creating balanced and representative datasets.

### ***2. Direct approach: Annotation during normal analysis***

Perfect for when you are already analyzing videos and decide to create a yolo model during the process.

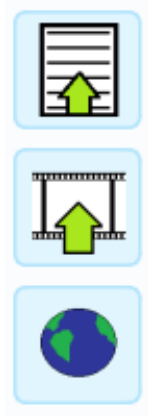
#### **Annotation in the main stream:**

- a. Upload your video to the platform as usual.
- b. Enable Manual Annotation during your routine analysis.
- c. Annotate organisms as they appear in the video (as you normally would).
- d. Final Processing: Export to YOLO (or export segmentation dataset): Saves the annotations (labels + images + dataset configuration file). Train YOLO Model (or segmentation model): Exports the annotations and automatically starts training the model.

**Advantages of this approach:**

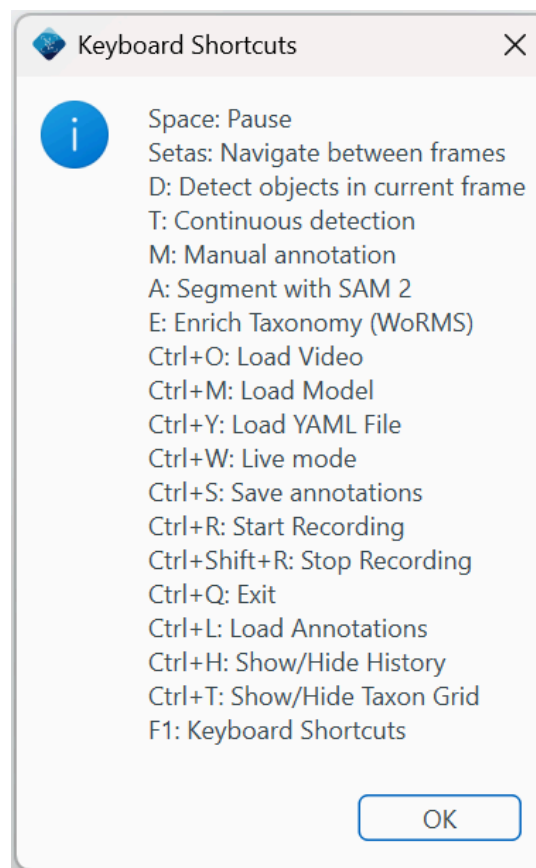
- a. Don't interrupt your normal workflow.
- b. Faster and more straightforward process.

## 6. Export and georeferencing



1. Export annotations in CSV format.
2. Export frames in PNG format.
3. Merge annotations CSV with georeferencing CSV.

## 7. Shortcuts



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For additional information or support send an email to: [raphaelaneves73@usp.br](mailto:raphaelaneves73@usp.br)

Repository: [GitHub - raoahela/iSEA](https://github.com/raoahela/iSEA)

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